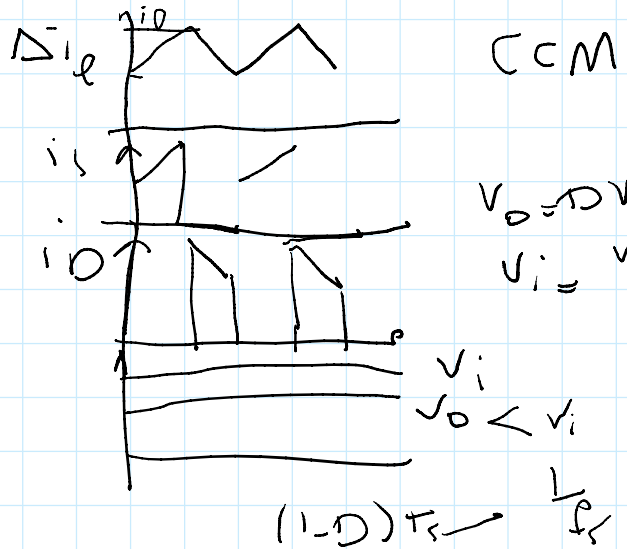
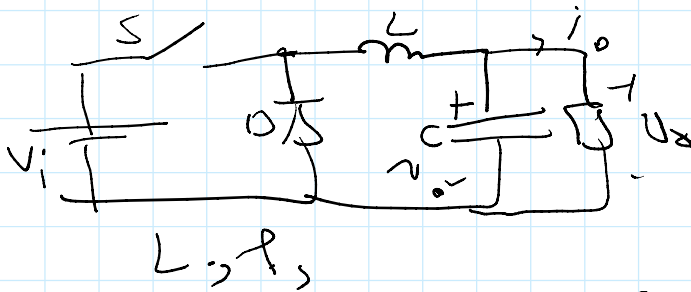


Recall switching power supplies
buck (basic) step-down chopper + LC filter



CCM: $\bar{i}_L \geq \frac{\Delta i_L}{2}$

switch off: $v_L = -V_o$

switch on

$$L \frac{\Delta i_L}{t_{off}} = V_o \Rightarrow \Delta i_L = \frac{V_o t_{off}}{L} = \frac{(1-D)V_o}{L f_s} \quad (1)$$

lossless

$$I_o = \frac{P_o}{V_o} = \frac{P_{in}}{V_o}$$

$\bar{i}_L, I_o, \bar{i}_{in} = D \bar{i}_L$

$$\bar{i}_L = \frac{\bar{i}_{in}}{D} = \frac{\frac{P_{in}}{V_i D}}{D} = \frac{P_o}{V_i D} = \frac{P_o}{V_o} = \frac{P_o}{V_o}$$

$$\frac{P_o}{V_o} \geq \frac{(1-D)V_o}{L f_s} \Rightarrow L \geq \frac{(1-D)^2 V_o^2}{P_{min} f_s} \quad \text{CCM}$$

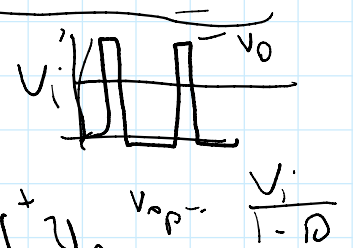
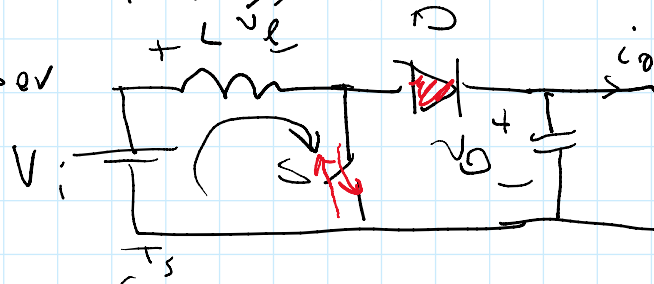
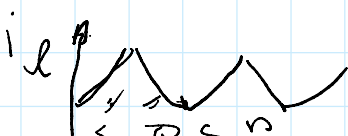
$V_i = V_{imin} - V_{max}$

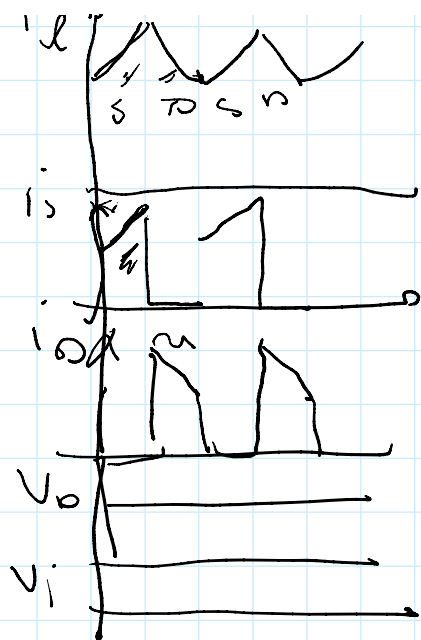
$$L \leq \frac{(1-D)^2 V_o^2}{P_{max} f_s} \quad \text{DCM}$$

$D: D_{min} - D_{max}$

Boost converter: $|V_o| \geq |V_i|$

step-up chopper





$$\int_0^{T_s} V_L dz = \text{net flux} = 0$$

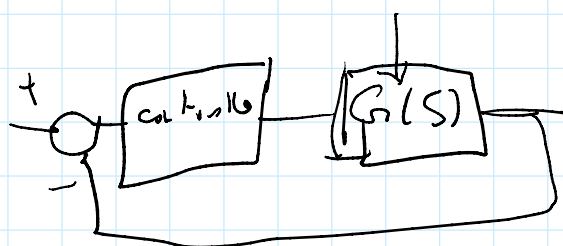
$$\int_0^{t_{on}} V_L dz + \int_{t_{on}}^{T_s} V_L dz = 0$$

$$V_i \times t_{on} + (V_i - V_o) t_{off} = 0$$

$$\frac{V_o}{V_i} = \frac{1}{1-D_s}, \quad M = \frac{1}{1-D}$$

$$\frac{I_o}{I_i} = 1 - D_s \quad \boxed{P_o = P_i}$$

lossless $D \rightarrow 1 \Rightarrow V_o \rightarrow \infty$

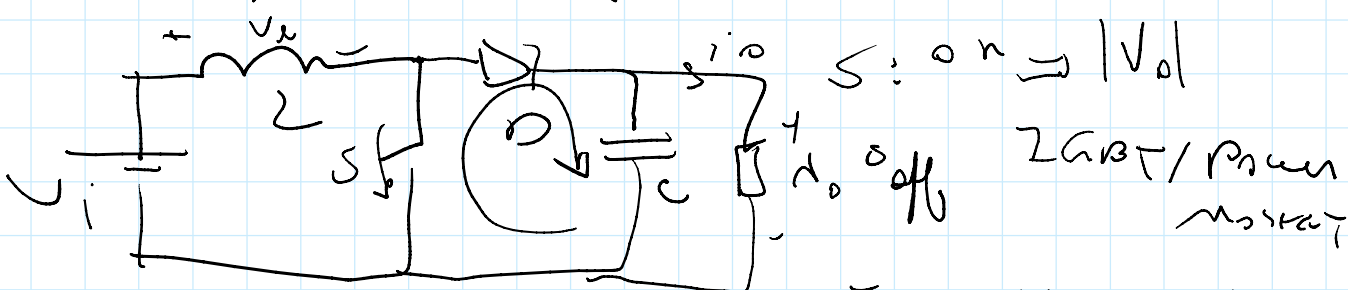


$$\begin{aligned} 0 < t < t_{on} & \left\{ \begin{aligned} V_i &= L \frac{di}{dt} \\ V_c &= V_o \end{aligned} \right. & \text{I} & \quad D = \frac{t_{on}}{T} \\ t_{on} < t < T & \left\{ \begin{aligned} L \frac{di}{dt} &= V_i - V_o \end{aligned} \right. & \text{II} & \quad \frac{t_{off}}{T} = 1-D \end{aligned}$$

State averaging

$$D \times \text{I} + (1-D) \times \text{II} \quad \checkmark$$

Q.M. $\rightarrow$ Non-minimum phase



$$V_{CE(sus)} \geq V_o \times S.F., \quad I_{CS} \geq I_s = D \bar{I}_L = D \bar{I}_{in}$$

$$\bar{I}_S = D \frac{P_{in}}{V_i} = D \frac{P_o}{V_i}, \quad I_S \geq P_{mon} \frac{P_{omar}}{V_{imin}}, \quad S.F.$$

$$V_{KEM} \geq V_o \times S.F.$$

$$I_D = (1-D) \bar{I}_L \quad \text{fast-recovery}$$

example: $V_i = 50-60V$
 $V_o = 100V$

$$\frac{1}{2} C V_{max}^2 - \frac{1}{2} C V_{min}^2 = P_{omar} \times t_{on}$$

example. $V_i = 50-60V$
 $V_o = 100V$
 $P_o = 1W = 100W$
 $\Delta V_o = 5\%$

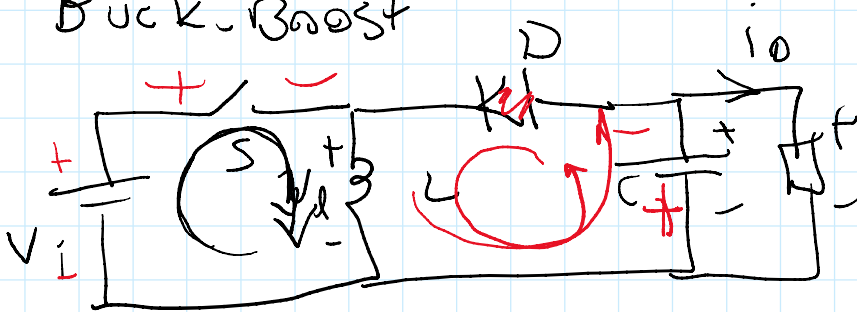
$$\frac{1}{2} C V_{max}^2 - \frac{1}{2} C V_{min}^2 = P_{out} t_{on}$$

$$\frac{V_o}{V_i} = \frac{1}{1-D} \Rightarrow$$

ccm $\textcircled{II} \bar{I}_L \geq \frac{\Delta i_L}{2}$, $\bar{I}_L = I_{line} = \frac{P_{in}}{V_{in}} = \frac{P_o}{V_o(1-D)}$ ✓ $\textcircled{2}$

(son) $L \frac{\Delta i_L}{T_{on}} = V_i \Rightarrow \Delta i_L = \frac{V_i t_{on}}{L} = \frac{V_o(1-D)D}{f_s}$ ✓ $\textcircled{I}$

Buck-Boost



$$\frac{V_o}{V_i} = \frac{-D}{1-D} \quad \left| \frac{V_o}{V_i} \right| = \frac{D}{1-D}$$

$$\left| \frac{I_o}{I_i} \right| = \frac{1-D}{D}$$

$$\int_{t_{on}}^{T_s} V_L dz = \text{Net Flux} = 0$$

$$V_o \int_{t_{on}}^{T_s} V_L dz + \int_{t_{on}}^{T_s} V_L dz = 0$$

$$V_i t_{on} + V_o t_{off} = 0$$

$0 < \frac{1}{2} - \text{buck}$
 $1 > \frac{1}{2} - \text{boost}$

3: IGBT/MOSFET

$$V_{ce(sus)} \geq (V_{if} + N_o) \propto s.p.$$

$$\bar{I}_c(s.p.) = \bar{I}_i \geq I_{max} \propto s.p. = \frac{P_{in}}{V_i} \propto s.p. = \frac{P_{out}}{V_{imin}} \propto s.p.$$

$$\Delta V_o = 5\% \Rightarrow \frac{1}{2} C V_{max}^2 - \frac{1}{2} C V_{min}^2 = P_{out} t_{on}$$

$$L: \bar{I}_L \geq \frac{\Delta i_L}{2} \quad L, f_s$$

